Unsupervised Learning: Validation & Workflow

Exploratory vs. Confirmatory Analysis

Confirmatory Analysis:

- Seeks to test an a priori hypothesis.
- Examples:
  - Classical inferential statistics.
  - Prediction in supervised learning.

Exploratory Analysis:

- Seeks to explore and understand data.
- Hypothesis generating.

Which is unsupervised analysis?

Unsupervised Learning Objectives

1. Data exploration.
   - Explore data to find patterns and generate hypotheses.
   - Exploratory analysis - Generated hypotheses will be later tested and confirmed.
2. Data-driven discoveries.
   - Make discoveries by finding important patterns and structures in the data.
   - Challenge: Both exploratory & confirmatory analysis!

Data Exploration

- Understand patterns, trends, and anomalies in your data.
- Prepare your data for the modeling stage and confirmatory analysis.
- Give a visual summary of the data.
- Generate hypotheses to be confirmed later.

We need to use multiple techniques to fully explore and visualize data!

From Data Exploration to Data-Driven Discoveries

Challenge:

- Any potential discoveries found during data exploration (hypothesis generation) cannot be confirmed on the same data set (hypothesis testing).

Why not?

From Data Exploration to Data-Driven Discoveries

Challenge:

- Any potential discoveries found during data exploration (hypothesis generation) cannot be confirmed on the same data set (hypothesis testing).

Why not?

Overfitting!

From Data Exploration to Data-Driven Discoveries

Challenge:

- Any potential discoveries found during data exploration (hypothesis generation) cannot be confirmed on the same data set (hypothesis testing).

Why not?

Overfitting!

What can we do to **validate** data-driven discoveries?

Validating Data-Driven Discoveries

1. Corroborate via existing literature.
    - Suggestive, but further confirmation needed.

Validating Data-Driven Discoveries

1. Corroborate via existing literature.
   - Suggestive, but further confirmation needed.
2. **Stability:** Show data-driven discovery is stable.
   - Small changes to the data, the algorithm, the method, the parameters, etc. yield the same or similar result.
   - Communicates uncertainty in discovery and avoids overfitting.

Validating Data-Driven Discoveries

1. Corroborate via existing literature.
   - Suggestive, but further confirmation needed.
2. **Stability:** Show data-driven discovery is stable.
   - Small changes to the data, the algorithm, the method, the parameters, etc. yield the same or similar result.
   - Communicates uncertainty in discovery and avoids overfitting.
3. **Generalizability:** Confirm via a completely separate test set.
   - Strong validation and standard in machine learning.
   - Sometimes challenging for unsupervised learning.

Validating Data-Driven Discoveries

1. Corroborate via existing literature.
   - Suggestive, but further confirmation needed.
2. **Stability:** Show data-driven discovery is stable.
   - Small changes to the data, the algorithm, the method, the parameters, etc. yield the same or similar result.
   - Communicates uncertainty in discovery and avoids overfitting.
3. **Generalizability:** Confirm via a completely separate test set.
   - Strong validation and standard in machine learning.
   - Sometimes challenging for unsupervised learning.
4. Validate via biological experiments.
   - Gold Standard! True confirmation.
   - Expensive & sometimes not possible.

Stability Principle

Idea:

- Reproducible / reliable discoveries are likely to be true discoveries.

Approach:

- Use repeated data perturbations of training data to mimic having new test data.
  - Subsampling, bootstrapping, random corruptions, random noise, random tuning parameters, random initializations, etc.
- Apply machine learning technique to each perturbed data set.
- If the same discovery appears repeatedly, then it's likely a true discovery.

Stability Principle

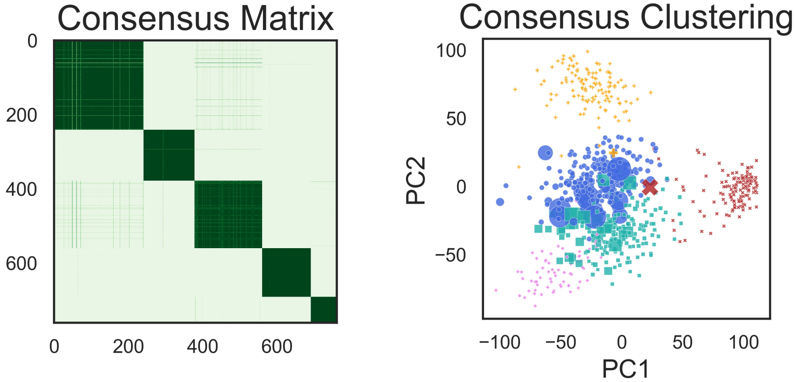
Strengths:

- Useful when a separate test set in unavailable or difficult to confirm via a test set.
- Communicates level of uncertainty associated with the data-driven discovery.
- Broadly applicable:
  - Consensus clustering.
  - Stable neighbors in dimension reduction.
  - Stable edge recovery in graph learning.

Weaknesses:

- Computationally more expensive.

Example: Consensus Clustering



PANCAN example from *Allen, Gan, & Zheng, 2024.*

Generalizability: Confirming Discoveries on a Test Set

Idea:

- Use a “training set” for data exploration and to make discoveries.
- Use a separate “test set” for confirming the discovery.

Approach:

- Use a separate study for a test set.
- Randomly split samples into a separate training and test set before any analysis.

Important:

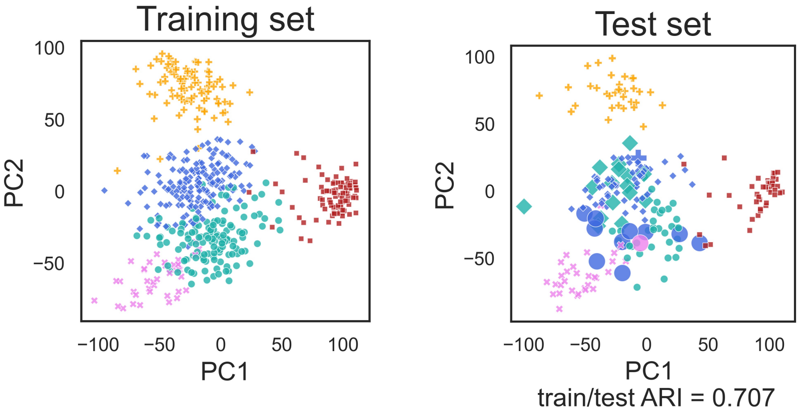
- Keep test set hidden until confirmatory stage!

But, this is often challenging for unsupervised learning . . . .

Example: Generalizability for Clustering

- Cluster the training data with K clusters.
- Separately, cluster the test data with K clusters.
- On training data, build a predictive model, $\hat{f}()$, to predict the K cluster labels.
  - E.g. KNN or RF.
- Use $\hat{f}()$ to predict labels on test set.
- Align predicted and cluster labels on test set (Hungarian algorithm).
- Use a metric to compare test set clusters to test set predicted labels.
  - E.g. Rand, Jaccard, or **Adjust Rand Index**.

Example: Generalizability for Clustering



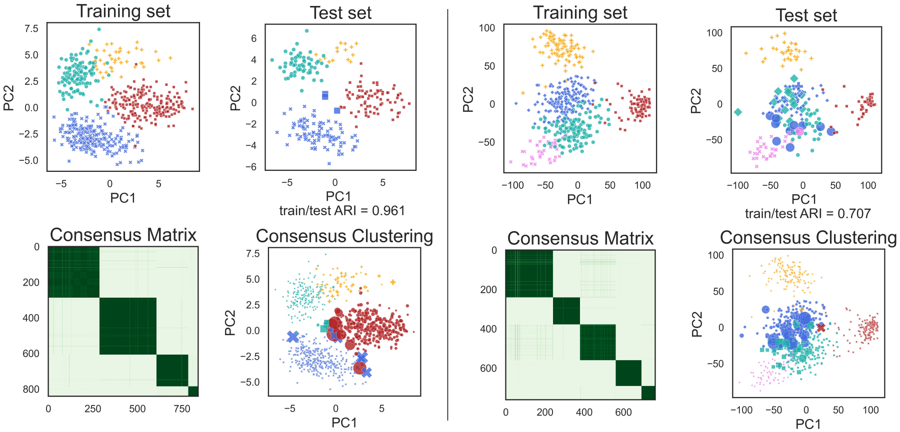
PANCAN example from *Allen, Gan, & Zheng, 2024.*

Discussion: Generalizability

Can we use generalizability to validate other unsupervised findings?

- Patterns from dimension reduction.
- Graphical models.
- Others?

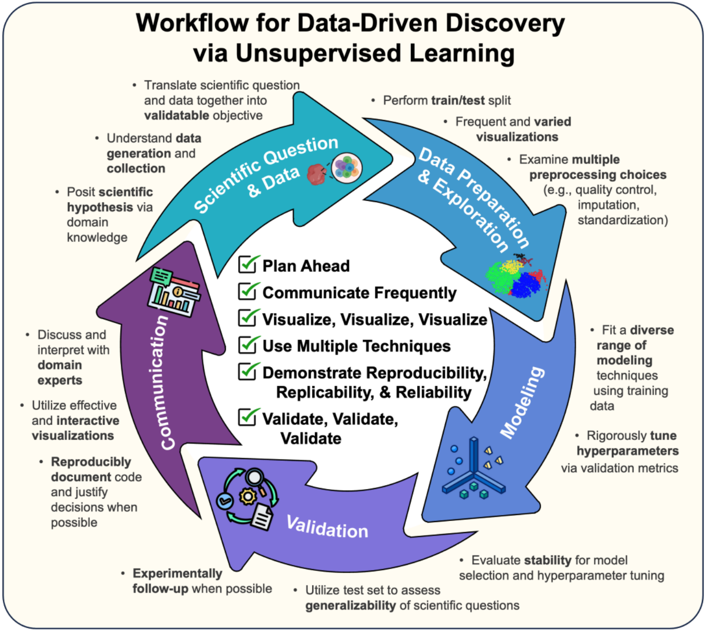
Cluster Validation Example



(a) Author (b) PANCAN

Allen, Gan, & Zheng, 2024.

Unsupervised Learning: Workflow & Best Practices



Chang, Tang, Zikry, and Allen, 2025

Workflow for Unsupervised Analysis

Scientific Question & Data:

- Posit scientific hypothesis based on domain knowledge.
- Understand data generation and collection.
- Translate scientific question and data together into a validate-able objective.
- Plan ahead!

Data Preparation & Exploration:

- Split data into training and test set.
- Explore frequent and varied visualizations.
- Examine multiple pre-processing choices and eventually sensitivity of results to pre-processing choices.

Workflow for Unsupervised Analysis

Unsupervised Modeling:

- Fit a diverse range of unsupervised techniques.
- Rigorously tune hyperparameters.

Validation:

- Evaluate stability of unsupervised results.
- Utilize test set to assess generalizability of results.
- Experimentally follow-up when possible.

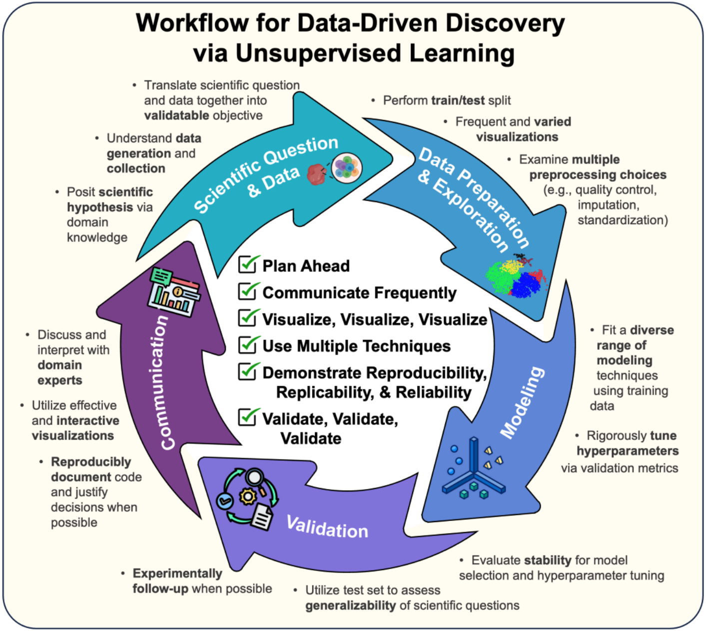
Workflow for Unsupervised Analysis

Communication:

- Discuss and interpret results with domain experts.
- Utilize effective and interactive visualizations.
- Communicate uncertainty.
- Reproducibly document code and justify analysis decisions.

Best Practices for Unsupervised Analysis

1. Plan Ahead.
2. Communicate Frequently.
3. Visualize, Visualize, Visualize.
4. Apply Multiple Techniques.
5. Validate, Validate, Validate.
6. Demonstrate Reproducibility, Replicability & Reliability.



Chang, Tang, Zikry, and Allen, 2025

Summary of This Course

Topics & Techniques covered:

- Dimension Reduction.
  - PCA.
- Pattern Recognition & Data Visualization.
  - PCA, NMF, ICA, MDS, tSNE, UMAP.
- Clustering.
  - K-means, Hierarchical, Spectral Clsutering, Modularity-Based Clustering.
- Feature Filtering via Association Testing.
  - FWER, FDR, Permutation approaches.
- Graphical Models.
  - Graph types, Gaussian graphical models.
- Validation.